LAB EXPERIMENT RECORD

Set 21 Energy_Consumption_Data.csv

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
Name	Swetha S
Register Number	192424049
Date	13/10/2026
Day	Tuesday

AIM

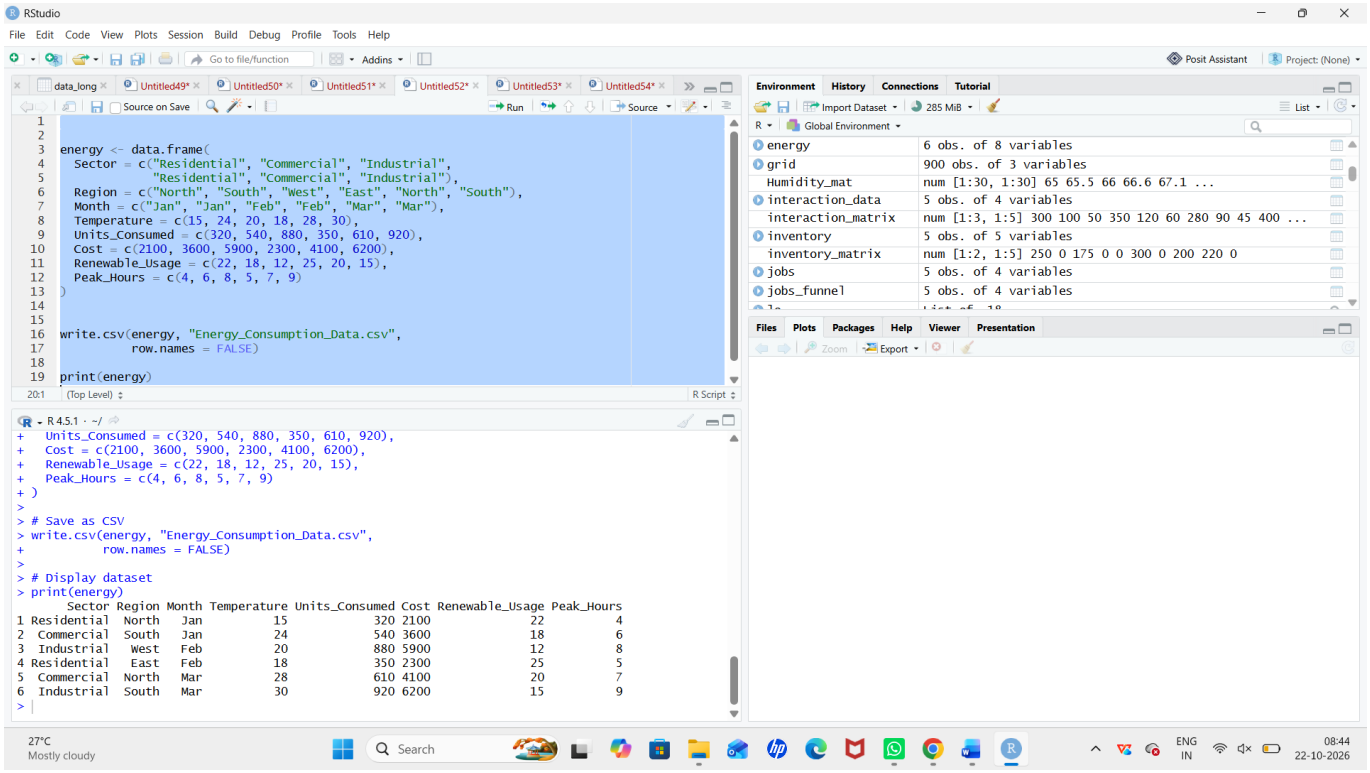
To analyze energy consumption data using R programming and create visualizations such as a histogram, density plot, scatter plot, and bar chart, and to develop an interactive Tableau dashboard showing monthly energy usage trends and sector-wise regional consumption.

1.

```
energy <- data.frame(
  Sector = c("Residential", "Commercial", "Industrial",
             "Residential", "Commercial", "Industrial"),
  Region = c("North", "South", "West", "East", "North", "South"),
  Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),
  Temperature = c(15, 24, 20, 18, 28, 30),
  Units_Consumed = c(320, 540, 880, 350, 610, 920),
  Cost = c(2100, 3600, 5900, 2300, 4100, 6200),
  Renewable_Usage = c(22, 18, 12, 25, 20, 15),
  Peak_Hours = c(4, 6, 8, 5, 7, 9)
)


write.csv(energy, "Energy_Consumption_Data.csv",
          row.names = FALSE)

print(energy)
```

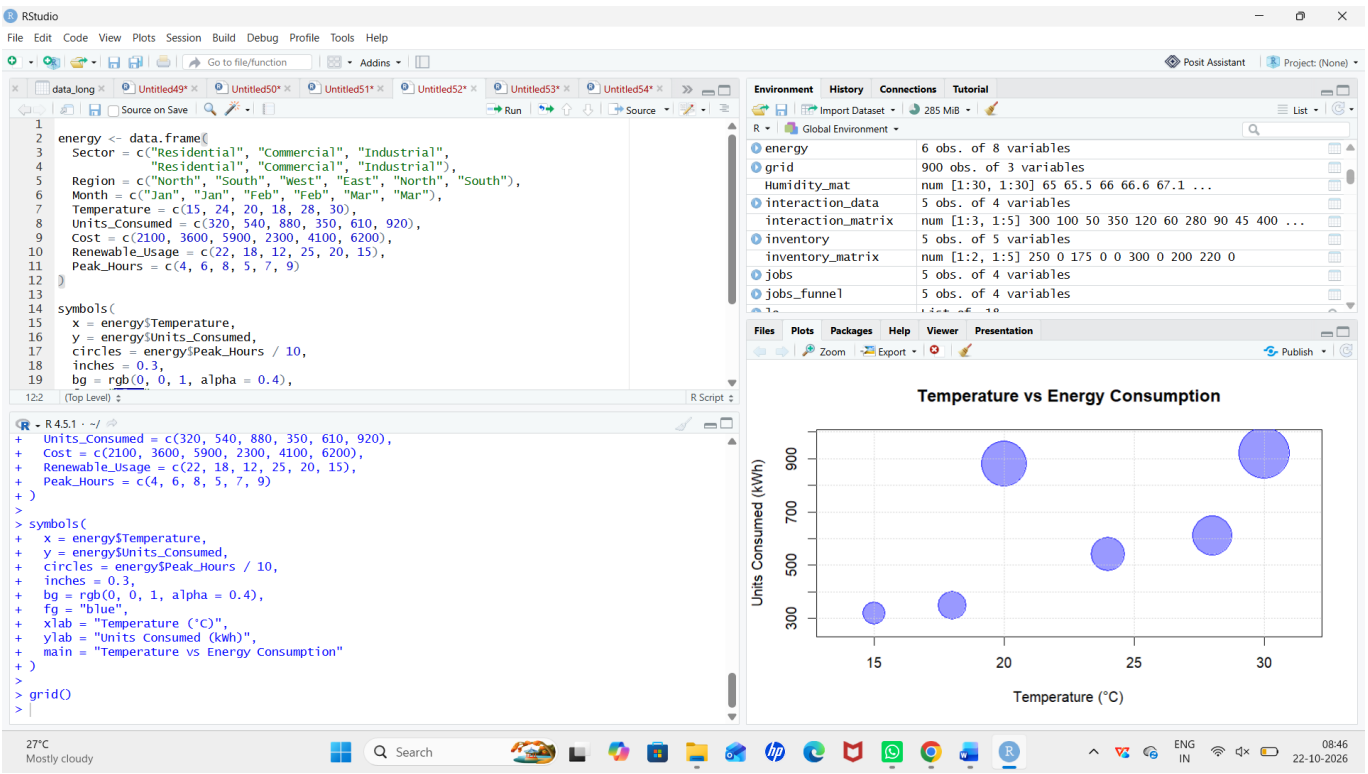


2

```
energy <- data.frame(
  Sector = c("Residential", "Commercial", "Industrial",
             "Residential", "Commercial", "Industrial"),
  Region = c("North", "South", "West", "East", "North", "South"),
  Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),
  Temperature = c(15, 24, 20, 18, 28, 30),
  Units_Consumed = c(320, 540, 880, 350, 610, 920),
  Cost = c(2100, 3600, 5900, 2300, 4100, 6200),
  Renewable_Usage = c(22, 18, 12, 25, 20, 15),
  Peak_Hours = c(4, 6, 8, 5, 7, 9)
)

symbols(
  x = energy$Temperature,
  y = energy$Units_Consumed,
  circles = energy$Peak_Hours / 10,
  inches = 0.3,
  bg = rgb(0, 0, 1, alpha = 0.4),
  fg = "blue",
  xlab = "Temperature (°C)",
  ylab = "Units Consumed (kWh)",
  main = "Temperature vs Energy Consumption"
)

grid()
```



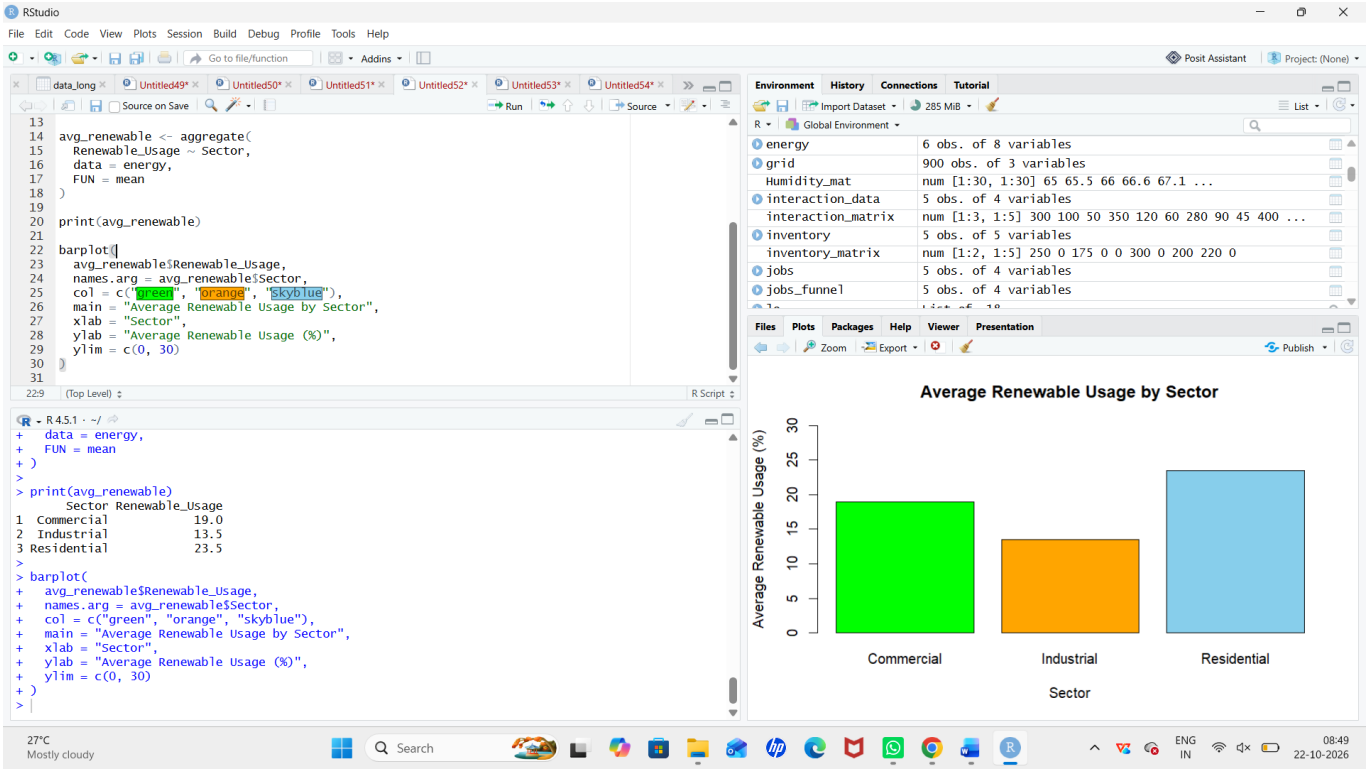
3.

```
energy <- data.frame(
  Sector = c("Residential", "Commercial", "Industrial",
             "Residential", "Commercial", "Industrial"),
  Region = c("North", "South", "West", "East", "North", "South"),
  Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),
  Temperature = c(15, 24, 20, 18, 28, 30),
  Units_Consumed = c(320, 540, 880, 350, 610, 920),
  Cost = c(2100, 3600, 5900, 2300, 4100, 6200),
  Renewable_Usage = c(22, 18, 12, 25, 20, 15),
  Peak_Hours = c(4, 6, 8, 5, 7, 9)
)

avg_renewable <- aggregate(
  Renewable_Usage ~ Sector,
  data = energy,
  FUN = mean
)

print(avg_renewable)

barplot(
  avg_renewable$Renewable_Usage,
  names.arg = avg_renewable$Sector,
  col = c("green", "orange", "skyblue"),
  main = "Average Renewable Usage by Sector",
  xlab = "Sector",
  ylab = "Average Renewable Usage (%)",
  ylim = c(0, 30)
)
```


RStudio
avg_renewable <- aggregate(
Renewable_Usage ~ Sector,
data = energy,
FUN = mean
)
print(avg_renewable)
barplot(
avg_renewable$Renewable_Usage,
names.arg = avg_renewable$Sector,
col = c("green", "orange", "skyblue"),
main = "Average Renewable Usage by Sector",
xlab = "Sector",
ylab = "Average Renewable Usage (%)",
ylim = c(0, 30)
)
Sector Renewable_Usage
1 Commercial 19.0
2 Industrial 13.5
3 Residential 23.5
Average Renewable Usage by Sector
Average Renewable Usage (%)
Commercial
Industrial
Residential
Sector

4.


Energy Consumption Dashboard
Analyze • Explore • Make Better Decisions
Filters
Month
All
Sector
All
Key Metrics
Total Units Consumed
3,170 kWh
Total Cost
₹ 20,000
Avg. Renewable Usage
17.5 %
Avg. Peak Hours
6.5 hrs
Monthly Energy Usage Trends
Units Consumed (kWh)
Month
Jan
Feb
Mar
Residential
Commercial
Industrial
Sector vs Region Heatmap (Units Consumed)
Sector
North
South
West
East
Residential
Commercial
Industrial
Units Consumed (kWh)

Procedure

- Create the energy consumption dataset containing sector, region, month, temperature, units consumed, cost, renewable usage, and peak hours.
- Import the required libraries and prepare the dataset in R.
- Generate a histogram and density plot to analyze the distribution of energy consumption.
- Create a scatter plot to study the relationship between temperature and energy consumption, using bubble size to represent peak hours.
- Calculate the average renewable energy usage for each sector and display it using a bar chart.
- Export the dataset as a CSV file and import it into Tableau.
- Create a line chart to visualize monthly energy usage trends.
- Create a heatmap to compare energy consumption across sectors and regions.
- Add Month and Sector filters and combine the visualizations into an interactive dashboard.

Result

The energy consumption data was analyzed successfully using R, and the visualizations helped identify consumption patterns, the relationship between temperature and energy usage, and average renewable energy usage by sector. An interactive Tableau dashboard was designed to display monthly energy trends, regional consumption, and key performance metrics.